Check for updates

The Dynamics of Thrombin Formation

Kenneth G. Mann, Saulius Butenas, Kathleen Brummel

Abstract—The central event of the hemostatic process is the generation of thrombin through the tissue factor pathway. This is a highly regulated, dynamic process in which thrombin itself plays many roles, positively and negatively its production and destruction. The hemostatic process is essential to normal physiology and is also the Achilles heel of our aging population. The inappropriate generation of thrombin may lead to vascular occlusion with the consequence of myocardial infarction, stroke, pulmonary embolism, or venous thrombosis. In this review, we summarize our present views regarding the tissue factor pathway by which thrombin is generated and the roles played by extrinsic and intrinsic factor Xa generating complexes in hemostasis and the roles of the stoichiometric and dynamic inhibitors that regulate thrombin generation. **(*Arterioscler Thromb Vasc Biol*. 2003;23:17-25.)**

Key Words: coagulation ■ fibrinogen ■ aggregation ■ coagulation inhibitors

The inventory of molecular components and the presumed physiology of the hemostatic process and its regulation have been established on the basis of plasma abundance, hemorrhagic or thrombotic pathology, in vitro tests, and chemical signatures. Two in vitro plasma tests, the prothrombin time and the activated partial thromboplastin time, were prominent in the development of the inventory. The former test relies on the addition of an extrinsic tissue factor source (thromboplastin), whereas the latter is based on the introduction of a foreign surface to initiate coagulation using only this surface contact and the biological constituents intrinsic to plasma. Both tests rely on a fibrin formation (clotting) end point. These assays permitted identification of connectivity between the component activities identified as required for plasma coagulation and defined the concept of intrinsic and extrinsic coagulation pathways, which converge at the step of formation of the prothrombinase complex. However, the mechanisms established by in vitro tests are not always mirrored in human pathology associated with bleeding or thrombosis. The primary pathway leading to hemostatic and thrombotic pathologies is associated with the tissue factor–initiated extrinsic coagulation pathway, whereas components unique to the intrinsic or contact pathway (factor XI, factor XII, prekallikrein, HMW kininogen) may have accessory roles in the process. Therefore, in this review, we focus on the dynamics of the reactions associated with the introduction of tissue factor to blood, leading to the formation of thrombin.

An evaluation of the reactions involved in the formation of thrombin leads to the conclusion that the physiologically relevant hemostatic mechanism is composed of 3 procoagulant vitamin K–dependent enzyme complexes (which use the proteases factor IXa, factor Xa, and factor VIIa) and one anticoagulant vitamin K–dependent complex.[1] Each complex involves a vitamin K–dependent serine protease and a cofactor protein with the protein-protein complex assembled on a membrane surface provided by activated or damaged cells. The same hemostatic process required for preventing leaks from the vasculature may also be life threatening when responsible for an intravascular occlusion. Thus, nature has elected a system for highly regulated, multiconstituent activity presentation that provides a process that will lead to the local arrest of hemorrhage. The plasma proteins involved in the process require activation to participate in the thrombin-generating process. In addition, platelet adhesion and activation are required to provide membrane binding sites explicitly at the region of vascular damage, because platelet adhesion and activation provides the discrete membrane sites on which all of the plasma-derived procoagulant complexes are assembled.[2–7] Equally important are the stoichiometric and dynamic inhibitory systems, which block the presentation of thrombin. The sum of inhibitory functions is far in excess of the potential procoagulant response. These inhibitory processes act in synergy, providing minimal activation thresholds, which must be achieved before significant thrombin generation.[8–10]

The significance of the components involved in the procoagulant response and its regulation in a genetically homogenous population is reflected in studies of function-deleted transgenic mice. In mice, elimination of tissue factor, factor VII, tissue factor pathway inhibitor (TFPI), factor X, factor V, prothrombin, and protein C is lethal, whereas deficiencies of factor VIII and factor IX contribute significantly to hemorrhagic risk.[11–19] It is instructive to note that in the outbred human population, individuals with equally severe prothrombin, factor X, factor V, and factor VII deficiencies do exist,

Received August 21, 2002; revision accepted October 22, 2002.

From the Department of Biochemistry, University of Vermont, College of Medicine, Burlington, Vt.

Correspondence to Kenneth G. Mann, Department of Biochemistry, 89 Beaumont Ave, Given Building, University of Vermont, Room C-401, College of Medicine, Burlington, VT 05405. E-mail kmann@zoo.uvm.edu

© 2003 American Heart Association, Inc.

Arterioscler Thromb Vasc Biol. is available at http://www.atvbaha.org DOI: 10.1161/01.ATV.0000046238.23903.FC

Downloaded from http://ahajournals.org by on March 25, 2026

and indeed most of these factors were discovered because of the presentation of a living propositus displaying hemorrhagic pathology.[20–25] Thus, genetic and environmental events can significantly alter a potentially lethal outcome. It is also interesting to note that congenital fibrinogen deficiency both in mice and man is not lethal and frequently only mildly symptomatic. Therefore, fibrin formation does not seem to be essential for survival.[26]

Overview

The key initiating event in the generation of thrombin depends on the interaction of membrane-bound tissue factor and factor VIIa, the latter of which is preexistent in the plasma milieu at ≈1% to 2% of the total factor VII concentration (10 nmol/L).[27,28] The source and presentation of active tissue factor is controversial.[29–33] However, its damage-related presentation is essential. The factor VII zymogen is cleaved at arginine 152 by a variety of proteases, including thrombin, factor IXa, factor Xa, and factor VIIa–tissue factor to produce the serine protease factor VIIa.[34] However, although this "enzyme" seems to possess all of the appropriate catalytic machinery to display the active site of an effective serine protease, it does not express proteolytic activity unless it is bound to tissue factor. Thus, naked factor VIIa at natural biological concentrations has no significant activity toward either factor IX or factor X before its binding to tissue factor.[35] The defective active site also makes factor VIIa impervious to the high concentration of antithrombin-III (AT-III) present in blood, which permits its continued existence.[36] The factor VIIa-tissue factor protein-protein interaction increases the k_{cat} of the enzyme for synthetic substrates by two orders of magnitude[37,38] and increases the rate of factor X activation by four orders of magnitude.[35,39] This latter increase is the result of the aforementioned improvement in catalytic efficiency and the membrane binding of the macromolecular substrates factor IX and factor X. Substrate-membrane binding leads to an effective reduction in K_m. Overall, the k_{cat} increase and K_m lowering leads to a 10^4 increase in the expression rate of the enzyme toward its natural substrates.

The factor VIIa-tissue factor (extrinsic) complex (Figure 1) catalyzes the activation of both factor IX and factor X, the latter initially being the more efficient substrate. Thus, the initial product formed by the extrinsic factor Xase is factor Xa. The factor IX zymogen is competitive with factor X and requires two peptide bond cleavages at arginines 145 and 180 for activity. Both of these cleavages are catalyzed by factor VIIa-tissue factor. Factor Xa, bound to a membrane, provides one of the two required cleavages (arginine145) to produce the intermediate factor IXα. Thus, this feedback cleavage by membrane-bound factor Xa enhances the rate of generation of factor IXa, which is completed with the second bond cleavage (arginine 180) by factor VIIa-tissue factor.[40]

The initial factor Xa produced when bound to a membrane activates tiny amounts of prothrombin to thrombin, albeit rather inefficiently; this initial thrombin is essential to the acceleration of the process by serving as the activator for platelets, factor V, and factor VIII.[41–43] Once factor VIIIa is formed, the factor IXa generated by factor VIIa-tissue factor

Figure 1. The 3 vitamin K–dependent procoagulant complexes. The thicker arrow associated with intrinsic factor Xase generation of factor Xa reflects the improved catalytic efficiency of this complex. Reprinted with permission from Mann KG. *Coagulation Explosion.* Burlington: Vermont Business Graphics; 1997.

combines with factor VIIIa on the activated platelet membrane to form the intrinsic factor Xase (Figure 1) that becomes the major activator of factor X. The factor VIIIa-factor IXa complex is (10^5- 10^6-fold more active than factor IXa alone and ≈50 times more efficient than factor VIIa-tissue factor in catalyzing factor X activation; thus, the bulk of factor Xa is ultimately produced by factor VIIIa-factor IXa.[40,44] Factor Xa combines with factor Va on the activated platelet membrane surface, and this "prothrombinase" catalyst (Figure 1) converts prothrombin to thrombin. Prothrombinase is 300 000-fold more active than factor Xa in catalyzing prothrombin activation.

The coagulation system is under extraordinarily tight regulation by both stoichiometric and dynamic inhibition systems. The tissue factor concentration threshold for reaction initiation is steep, and the ultimate amount of thrombin produced is largely regulated by the stoichiometric inhibitors tissue factor pathway inhibitor (TFPI) and antithrombin III (AT-III) and by the dynamic anticoagulant process of protein C activation and functional expression (Figure 2).[9,10,45–51]

The principal influence of TFPI is to block the tissue factor-factor VIIa-factor Xa product complex, thus effectively neutralizing the extrinsic factor Xase and eliminating this catalyst's generation of both factor Xa and factor IXa.[46] TFPI is present in a low abundance (≈2.5 nmol/L) in blood; it is also releasable from the vasculature by the action of heparin.[52]

The stoichiometric inhibitor AT-III is normally present in plasma at more than twice the concentration (3.4 μmol/L) of any potential target coagulation enzyme generated by the tissue factor pathway. AT-III is an effective neutralizer of all of the procoagulant serine proteases.[45] The targets of AT-III are primarily the mature enzyme products of these reactions. AT-III is a weaker inhibitor of the factor VIIa-tissue factor complex[36]; thus, the principal influence of this inhibitor is in quenching thrombin production and the thrombin produced.

Downloaded from http://ahajournals.org by on March 25, 2026

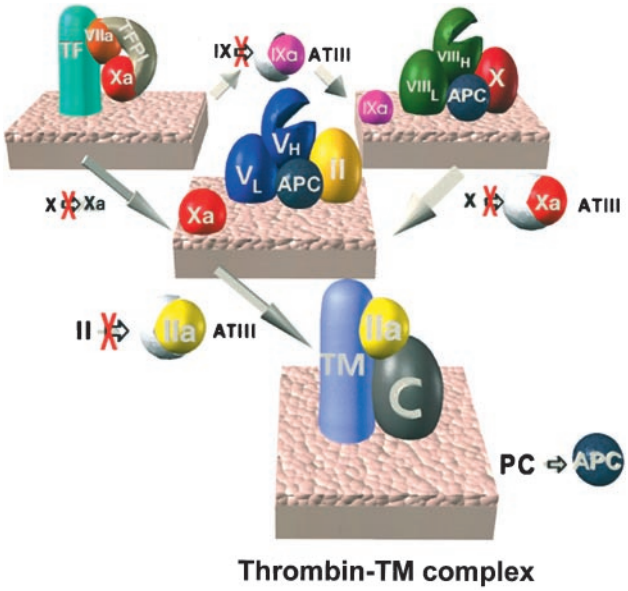

Figure 2. The procoagulant and anticoagulant vitamin K–dependent complexes and regulators. Antithrombin III neutralizes all the serine proteases present in the reaction mixture with the exception of factor VIIa. The tissue factor-factor VIIa complex is neutralized by the tissue factor pathway inhibitor as the factor Xa product complex. Thrombin binds to thrombomodulin and activates protein C. APC competes with factor Xa and factor IXa for factor Va and factor VIIIa binding and ultimately leads to inactivation of factor Va and factor VIIIa by proteolytic cleavage. Reprinted with permission from Mann KG. *Coagulation Explosion.* Burlington: Vermont Business Graphics; 1997.

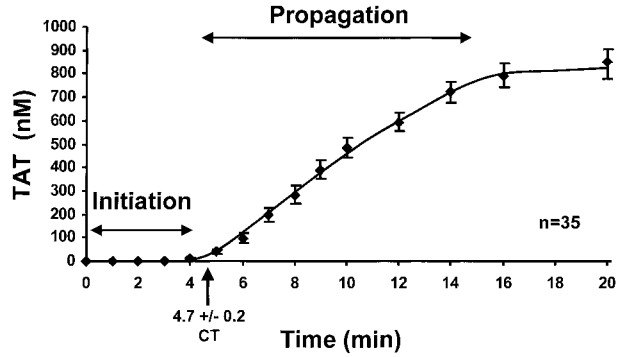

Figure 3. A summary of 35 paravivo experiments showing the generation of thrombin-antithrombin III complex (TAT) after the addition of 5 pmol/L tissue factor, 25 nmol/L phospholipid in contact pathway suppressed whole blood at 37°. The average clotting time±SEM (CT) is illustrated under the horizontal axis. From: Brummel KE, Paradis SG, Butenas S, Mann KG. Thrombin functions during tissue factor-induced blood coagulation. *Blood*. 2002;100:148–152. Copyright American Society of Hematology, used by permission.

To initiate the dynamic protein C system, the product enzyme thrombin binds to constitutively present vascular thrombomodulin (Tm) and activates the protein C (PC) to its activated species APC.[48] APC competitively binds with both factor VIIIa and factor Va, interfering with the formation of the "prothrombinase" and the "intrinsic Xase," initially by competition with factor Xa and factor IXa and ultimately by cleaving the cofactor factor Va and factor VIIIa to eliminate these complexes.[10,49–51,53,54]

Models of the Blood Coagulation Reaction

Our laboratory has made use of 4 models in attempts to recapitulate the dynamic mechanism of the coagulation reaction system. These have included the following: (1) synthetic plasma, in which mixtures prepared from purified procoagulants and anticoagulants and a natural or synthetic membrane source are induced to react by the addition of lipid reconstituted tissue factor[9,10,41,55–60]; (2) paravivo studies of coagulation, in which whole, contact pathway–suppressed blood obtained by phlebotomy and maintained at 37° is induced to clot by addition of a fixed amount of membrane reconstituted tissue factor[42,61–68]; (3) numerical (computer) models of the coagulation system, which are based on the ensemble of published (and estimated) rate constants and concentrations of procoagulants and stoichiometric inhibitors and the mechanisms of their interactions[69–71] (also referred to as "in silico" simulations); and (4) in vivo studies of the hemostatic reaction, in which whole blood exuding from a microvascular wound is sequentially sampled for relevant product formation.[72–75]

Each of the model systems used has specific benefits and limitations. The in vivo model, involving a "Simplate" wound, is the least flexible, involves the highest degree of discomfort for the volunteer, and is analytically difficult; however it is the most biologically relevant. The numerical model is the least expensive and the most rapid and convenient method of analysis. It provides insight into the regulation of reaction mechanisms occurring at concentrations of intermediates and products, which may be inferred but not measured with existing technology. It is (obviously) the least biologically relevant.

Thrombin Generation

Regardless of the model system chosen, the display of thrombin generation after tissue factor initiation of the hemostatic reaction is approximately the same.[41,42,56,71,73] This behavior is illustrated in Figure 3, which shows the generation of the thrombin-AT-III complex (TAT) as a function of time using the paravivo model. From an operational perspective, thrombin generation may be described as occurring in two phases. Shortly after the addition of tissue factor, tiny amounts of thrombin (nanomolar) are produced in an interval, which we define as the initiation phase of the reaction. Subsequently, the major bolus (>96%) of thrombin is produced during the propagation phase of the reaction. During the initiation phase of the reaction, the factor VIIa-TF complex forms and generates small amounts (subpicomolar) of factor Xa and factor IXa.[40,41,55] Factor Xa in collaboration with the membrane surface activates a small amount of prothrombin to thrombin, which serves to generate the platelet membrane and cofactor components required for the major generation of thrombin.

The early events associated with thrombin function during the initiation phase of the reaction are illustrated in Figure 4, which shows the inception points for the detection of thrombin products generated during the reaction measured in

Downloaded from http://ahajournals.org by on March 25, 2026

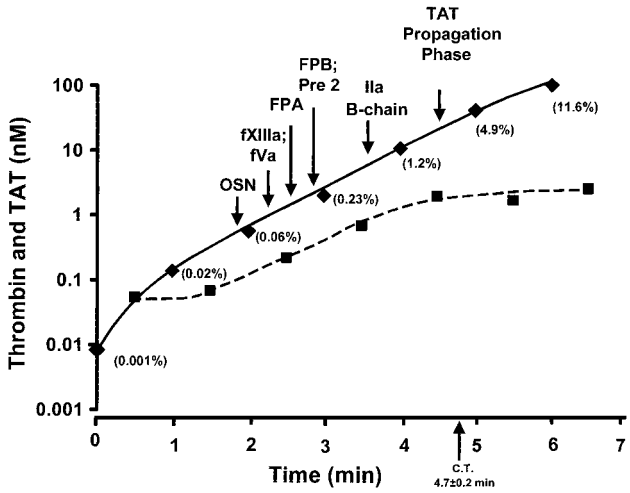

Figure 4. The generation of thrombin activation products during the initiation phase of the reactions shown in Figure 3. Platelet activation (osteonectin release) is detected at <1 nmol/L thrombin (0.06% of the total thrombin ultimately produced). The clotting time, 4.7 minutes, is observed immediately after the inception of the propagation phase of thrombin generation. The dotted line (■) corresponds with the active thrombin present in the reaction mixture, as determined from fibrinopeptide release. From: Brummel KE, Paradis SG, Butenas S, Mann KG. Thrombin functions during tissue factor-induced blood coagulation. *Blood*. 2002;100:148–152. Copyright American Society of Hematology, used by permission.

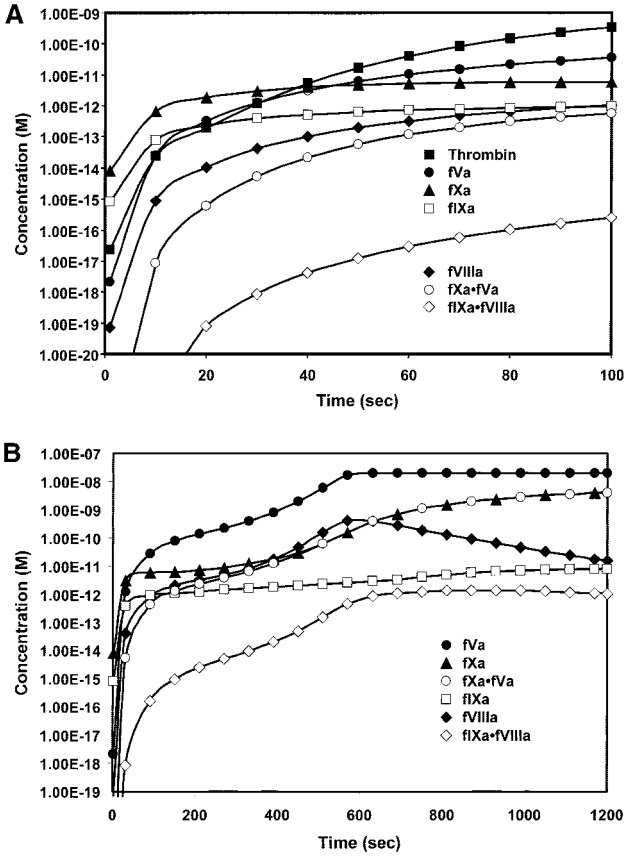

Figure 5. A, Numerical model of the generation of elements of the procoagulant pathway during the first 100 seconds of an in silico reaction stimulated by 5 pmol/L tissue factor. B, Computer representation of product formation during the 20-minute interval of the in silico reaction stimulated by 5 pmol/L tissue factor. Adapted with permission from Hockin MF, Jones KC, Everse SJ, Mann KG. A model for the stoichiometric regulation of blood coagulation. *J Biol Chem*. 2002;277:18322–18333.

paravivo experiments.[42] Many of these products are required to provide the catalysts (Figure 1) that generate most of the thrombin produced during the propagation phase of the reaction.

Under normal circumstances, the rate-limiting component of most prothrombinase formation and the generation of thrombin activity is the concentration of factor Xa.[41,55,61] Thus, under normal conditions, the activation of factor V and the activation of platelets (probably through thrombin-PAR-1 receptor interactions) occur rapidly to produce surplus factor Va and platelet membrane binding sites.[61] However, under conditions of congenital deficiency, thrombocytopenia, platelet pathology, or pharmacologic intervention, the tissue factor initiated reaction can become sensitive to factor V or platelets.[64,66,74] In studies to assess the influence of preactivation of platelets, we used the thrombin receptor activation peptide to provide for the preexpression of complex binding sites on the platelet surface.[66] Preactivation of the platelets did not change the rate of thrombin generation. However, pharmacologic interventions with agents such as PGE_1, ReoPro (Abciximab), and Integrilin do influence the generation of thrombin in the reaction.[66] Similarly, reductions in platelet counts (<10 000/mm^3) elicit dependence on platelet concentration.[64]

The end point used in evaluating hemostasis in most bioassays is the generation of a fibrin clot. As illustrated in Figure 4, the formation of a fibrin clot occurs at 10 to 30 nmol/L thrombin or ≈3% of the total amount of thrombin produced during the reaction, which is provided by only ≈7 pmol/L prothrombinase.[61] Thus, most thrombin formation is ignored using present technology for evaluating clinical hemorrhagic risk or thrombosis.

Reactant Evolution

Figures 5A and 5B represent the numerical model's evaluation of product formation over the entire time course of the reaction illustrated in Figure 3.[71] Panel 5A represents the first 100 seconds of the reaction and corresponds primarily to the initiation phase of the reaction. Note that the vertical axis is exponential. The initial burst of factor Xa (≈10 fmol/L) (▲), primarily generated by factor VIIa-TF, produces a small amount of thrombin (<10 fmol/L) (■), which activates some factor V and factor VIII to factor Va (●) and factor VIIIa (◆) and leads to the initial formation of the intrinsic factor Xase (◇) and prothrombinase (○) complexes. During this early phase of the reaction, thrombin generation is principally under the control of factor VIIa-tissue factor and factor Xa-membrane and the reaction is principally regulated by TFPI.

Figure 5B represents the entire reaction (20 minutes). Note that the formation of the prothrombinase complex (○) is identical to and ultimately limited by the concentration of factor Xa[55,61] (▲), because factor Va (●) is in excess. Also apparent is that the existence of factor VIIIa-factor IXa (the intrinsic factor Xase) (◇) is dependent not only on the

Downloaded from http://ahajournals.org by on March 25, 2026

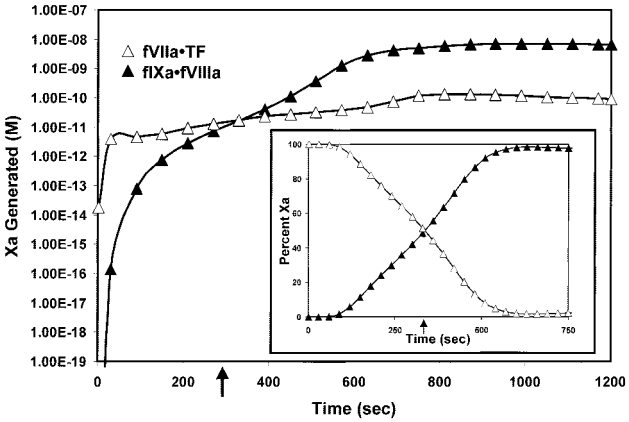

Figure 6. The contributions of the intrinsic and extrinsic factor Xase to factor Xa generation over the time course of the reaction. The inset illustrates the relative percentages of factor Xa generated by each catalyst. Reprinted with permission from Hockin MF, Jones KC, Everse SJ, Mann KG. A model for the stoichiometric regulation of blood coagulation. *J Biol Chem*. 2002;277:18322–18333.

stoichiometric concentrations of factor VIIIa (◆) and factor IXa (□) but also the dissociation constant between these two proteins and the instability of factor VIIIa, which spontaneously dissociates, limiting the concentration of factor VIIIa.

Significance of Intrinsic Factor Xase

The relative factor Xa generation by the factor VIIa-tissue factor and the factor IXa-factor VIIIa complexes is illustrated in Figure 6.[71] Initially, the concentration of the factor VIIa-tissue factor complex ($\approx 10^{-11}$ mol/L) is higher than the concentration of the factor VIIIa-factor IXa complex, which requires activation and assembly. As time progresses, however, the contribution of the latter, more active complex in factor Xa generation exceeds that of the extrinsic factor Xase. As a consequence, most of factor Xa (Figure 6 inset) is ultimately produced by the factor VIIIa-factor IXa complex in the tissue factor–initiated hemostatic processes. Clotting would have occurred at 250 to 300 seconds at this tissue factor concentration.

In the absence of factor VIII or factor IX, the intrinsic factor Xase cannot be assembled; thus no propagation phase occurs. This is the principal defect observed in hemophilia A and hemophilia B[62,67] in all models, including paravivo blood studies from those affected with these hemorrhagic abnormalities.

Because the presentation of a clot depends only on the generation of 10 to 30 nmol/L thrombin, at high tissue factor concentration, robust generation of factor Xa by factor VIIa-tissue factor can completely mask the contribution of the factor VIIIa-factor IXa complex in clot end point assays. This is the case for the prothrombin time in which the concentrations of thromboplastin (tissue factor and phospholipid) are chosen to produce a clot time of 11 to 15 seconds. This corresponds to a tissue factor concentration of >20 nmol/L. For the illustrations of Figures 3 through 6, a concentration of 5 pmol/L TF was used, producing a clotting time of ≈5 minutes. In hemophilia A and B, at these TF concentrations, although the clotting time is prolonged, the major defect is associated with the absence of a propagation phase.[62,67]

Attenuation of Thrombin Generation

The attenuation of the coagulation system is as important as the procoagulant process and involves both stoichiometric and dynamic regulators. TFPI and AT-III are the principal stoichiometric inhibitors of the process. TFPI is the principal regulator of the initiation phase of thrombin generation, whereas AT-III serves to attenuate thrombin activity and its generation.[9] These two agents when combined provide a synergistic regulatory effect by inducing kinetic thresholds such that the initiating tissue factor stimulus must be of significant magnitude to propel thrombin generation. Tissue factor concentrations below the threshold concentration required are ineffective in promoting massive thrombin generation because of the cooperative influence of the inhibitors; concentrations in excess of the threshold yield robust and almost equivalent thrombin generation.[9,71] In a similar fashion, TFPI and the PC-Tm-thrombin APC system cooperate to provide a similar, threshold-limited, synergistic inhibition of thrombin production.[10] The dynamic activated protein C system is responsive only after thrombin has been generated, because this system depends on activation of the zymogen protein C by the thrombin produced in the procoagulant response. Thus, its influence is mostly associated with quenching the propagation phase of thrombin generation,[10] although the two phases are significantly overlapped.

The activations of the cofactors factor V and factor VIII are multistep processes, and at high thrombomodulin concentrations, the protein C system is an effective neutralizer of the reaction. Factor V activation involves cleavages at arginines 709, 1018, and 1545. The cleavage at arginine 709 occurs first and produces the heavy chain (residues 1 through 709) of the molecule. Factor V activity, however, requires the cleavage at arginine 1545 to produce the light chain (residues 1546 through 2329) of factor Va.[76–79] APC inactivates factor Va (and intermediates in the activation process) principally by cleavage at arginines 506 and 306. Each of these cleavages is in the heavy chain.[53] Thus, the heavy chain can be inactivated before generation of the light chain of factor Va, eliminating its procoagulant activity.[56]

Accessory Processes

The zymogen factor XI is a symmetrical two-chain serine protease precursor present in plasma and platelets that has been variably associated with hemorrhagic pathology.[80–84] The significance of factor XI as an important procoagulant is established by the bleeding pathology associated with its qualitative or quantitative absence. This zymogen is also a substrate for thrombin and has been invoked in a revised pathway of coagulation.[85] Paravivo studies of the clotting of natural hemophilia C blood and synthetic plasma experiments, which mimic factor XI deficiency, illustrate the importance of the feedback activation of factor XI but only at the lowest TF concentrations.[62,86] At moderate concentrations of tissue factor (5 to 10 pmol/L), which produce clotting times in the range of 3 to 5 minutes, factor XI has little or no effect on thrombin generation or other procoagulant param-

Downloaded from http://ahajournals.org by on March 25, 2026

eters. However, at lower levels of tissue factor (1 to 2 pmol/L), which produce clotting in the range of 12 to 15 minutes, the generation of thrombin and formation of fibrin are prolonged in factor XI deficiency.[62] The variability of observations of pathology with factor XI deficiency is most likely a reflection of the dimension of the stimulus associated with the vascular lesion in factor XI–deficient individuals.

The platelet also provides a source of factor XI and factor V, both of which are secreted during the platelet activation process.[87–91] In studies of synthetic systems with deletion of plasma factor XI and factor V in which the platelet is the sole contributor of factor V and factor XI, in vitro data would suggest that the contributions of factor V or factor XI from platelets are sufficient and at least equivalent to the contribution of factor V or factor XI from plasma.[64] Conversely, plasma factor V and plasma factor XI seem to make the platelet contribution superfluous. These in vitro studies, however, are not entirely born out by in vivo observations of individuals who lack platelet factor V and have a bleeding diathesis.[92] Pathology does not seem to be associated with factor XII, high molecular weight kininogen, or prekallikrein deficiency states. These zymogens are cleaved to their active species by thrombin and thus may participate in coagulation but not as primary initiators of the hemostatic response.

Fibrin and Fibrinolysis

In purified systems, the kinetics of fibrinogen cleavage by thrombin result in the sequential removal of fibrinopeptide A (FPA) followed by fibrinopeptide B (FPB). The removal of FPA permits end-to-end polymerization, whereas removal of FPB has been thought a requirement for side-to-side polymerization,[93,94] with the latter required for covalent cross-linking, ie, the formation of isopeptides between ϵ-aminolysyl and γ-glutamyl residues of adjacent chains.[95] Elegant x-ray crystallographic studies of fibrinogen fragments provide details of these associations at the atomic level.[96] Although the processes have been well established in purified systems, fibrin formation during blood clotting per se has only recently been evaluated using the paravivo system.

During the tissue factor–induced coagulation of whole blood modified only by suppression of the contact pathway, the activation of factor XIII and the removal of FPA by thrombin occur simultaneously.[63] Cross-linked products are observed in the solution phase of the blood before the observation of a clot. At the point of clotting, $<$50% of the Aα chains of fibrinogen are cleaved and there is virtually complete removal of all fibrinogen/fibrin from solution. Thus, one must conclude that the initial clot is a heteropolymer, most likely composed of fibrinogen, fibrin 1, and partially cleaved fibrin 1. Subsequent to clotting, one begins to observe FPB in solution. However, this process never goes to completion, and only $\approx$40% of FPB is ever removed and is detectable in the fibrin clot as the Bβ chain of fibrin 1. In the initial clot, the products associated with γ-γ dimerization and α-chain cross-linked polymers are observed. The FPB, which is released into solution, is additionally processed by a carboxypeptidase B-like enzyme, which removes the $-$COOH terminal arginyl residue.[63] A prospective candidate of this reaction is the thrombin activatable fibrinolysis inhibitor (TAFI), which is activated by thrombin-Tm to the product TAFIa.[97,98] TAFIa cleaves terminal lysyl and arginyl residues from the partial plasmin digestion products of fibrin. The result is indirect stabilization of the clot, because clot lysis is accelerated by the relatively high affinity of plasmin for substrates terminating in lysine, thus accelerating plasminogen activation.[99]

Pharmacological and Genetic Alterations

Observations from fundamental kinetic studies can be translated back to considerations of the pathophysiology of hemorrhagic and thrombotic disease. Hemorrhagic syndromes are characterized as severe, moderate, or mild on the basis of the relative functional level of a coagulation factor. Severe hemophilia A is associated with factor VIII concentrations $<$1%.[80] Similar observations have been made for other coagulation factor deficiencies and provide paradigms for management of replacement therapy. These subjective assessments derived from clinical experience are consequences of the effective plasma protein concentrations required to generate effective hemostasis. In contrast to expectations, which may be derived from steady-state kinetics, in the pre–steady-state processes relevant to biology, only tiny amounts of the procoagulants upstream from prothrombinase are consumed during the thrombin-generating reaction, and contributions to rate processes rather than stoichiometric considerations predominate.[41,55,61,71]

The pathophysiologic influence of qualitative and quantitative alterations to the hemostatic system reactants has primarily been explored at the extremes of population distribution curves. These extremes are associated with congenital hemophilias and thrombophilias. In most cases, diagnostic methods are based on evaluations of single analytes with respect to their anticipated influence on the generation of hemorrhagic or thrombotic pathology. Synthetic plasma, numerical, and paravivo analyses show that quantitative differences in coagulation factor levels within the normal range, when combined, can produce significant variability in response to a tissue factor stimulus.[59,68] This sort of behavior, as illustrated for prothrombin concentrations in synthetic plasma and numerical models, is presented in Figure 7.[71] This figure displays the active thrombin concentration evaluated using the synthetic plasma and numerical systems as a function of time for a variety of prothrombin concentrations ranging from 50% to 150% of the population mean value. It can be seen that the active thrombin concentration present at any point in time as well as the cumulative amount of thrombin produced is significantly different for values of prothrombin concentration, which remain within the normal range. The stimulus-response algorithm that is presented for individuals with different prothrombin concentrations may be significant for predicting hemorrhagic or thrombotic risk, especially when such states are combined with other excursions from mean values. In the future, it may be anticipated that algorithms that combine individual coagulation factor and inhibitor levels will be useful in diagnosis and selection of prophylactic regimens.

The paravivo and in vivo systems have been used to evaluate pharmacologic interventions used to treat hemor-

Downloaded from http://ahajournals.org by on March 25, 2026

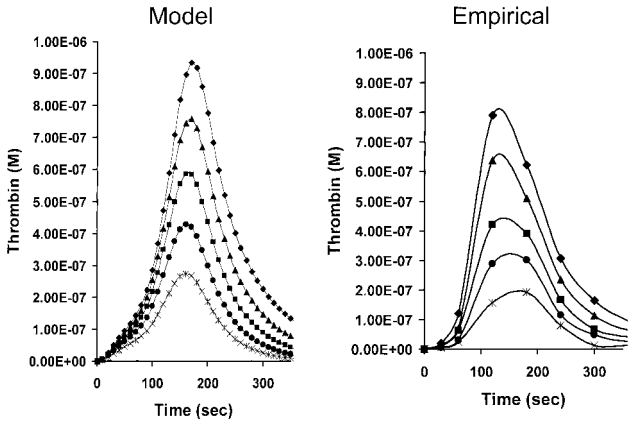

Figure 7. The concentration of active thrombin as a function of time in the presence of prothrombin at 150% (diamonds), 125% (triangles), 100% (squares), 75% (circles), and 50% (asterisk) of its mean physiological concentration. Represented are the empirical experiments of Butenas et al[59] and an in silico representation of the same experiment. Reprinted with permission from Hockin MF, Jones KC, Everse SJ, Mann KG. A model for the stoichiometric regulation of blood coagulation. *J Biol Chem*. 2002;277:18322–18333.

rhagic and thrombotic diseases, to evaluate congenital abnormalities, and to inspect the influence of common polymorphisms. Studies have been conducted on individuals with hemophilia A, B, and C, factor VII deficiency, the Pl^{A1}, Pl^{A2} platelet receptor polymorphism, and the factor XIII Val34-Leu polymorphism.[62,67,74,75] Pharmacological interventions have included ingestion of aspirin, Simvastatin, and Coumadin anticoagulation.[66,68,72–74] PGE_1, dipyridamole (Persantine), ReoPro (Abciximab), and Integrilin have been added in vitro to paravivo experiments.[66] In most instances, pharmacologic agents and abnormalities with an adverse effect associated with hemorrhage risk or a positive influence with respect to protection from thrombosis slightly prolong the initiation phase of the reaction but have their most dramatic influences on the propagation phase, when most thrombin is generated. As a consequence, clot end point–based assays for evaluation of hemorrhage risk or antithrombotic effect frequently do not display prolongations of the fibrin clotting time with these agents.

Summary

Technical advances in molecular genetics, protein chemistry, bioinformatics, and physical biochemistry provide us with an impressive array of tools and information with respect to normal and pathologic processes leading to hemorrhagic or thrombotic disease. The challenge for the 21st century will be to merge these mechanism-based, quantitative data sets with epidemiologic studies and clinical experience associated with the tendency to bleed or thrombose and with the therapeutic management of individuals with thrombotic or hemorrhagic disease. Our knowledge of the biology of coagulation is incomplete without considerations of genetics, biochemistry, pathology, hematology, and vascular biology. Because the composite of biochemical models needs to become biologically relevant and biological approaches need to become quantitatively enabled, we need to distill and integrate mechanistic data with the vast amount of subjective clinical experience regarding the management of individuals with thrombotic and hemorrhagic disease. Algorithms must be developed that can combine the art of clinical management with the quantitative science available in this field to define the outcome of the challenge or the efficacy of an intervention. Ultimately, we must tailor diagnosis and pharmacologic intervention to the individual.

Acknowledgements

This work was supported by grants HL-4670, HL-34575, and Training Grant HL-07594 from the National Institutes of Health.

References

1. Jenny NS, Mann KG. Coagulation cascade: an overview. In: Loscalzo J, Schafer AI, eds. *Thrombosis and Hemorrhage*. Baltimore, Md: Williams and Wilkins; 1998:3–27.
2. Rosing J, van Rijn JLML, Bevers EM, van Dieijen G, Comfurius P, Zwaal RF. The role of activated human platelets in prothrombin and factor X activation. *Blood*. 1985;65:319–332.
3. Tans G, Rosing J, Thomassen MC, Heeb MJ, Zwaal RF, Griffin JH. Comparison of anticoagulant and procoagulant activities of stimulated platelets and platelet-derived microparticles. *Blood*. 1991;77:2641–2648.
4. Bevers EM, Comfurius P, Zwaal RF. Changes in membrane phospholipid distribution during platelet activation. *Biochim Biophys Acta*. 1983;736: 57–66.
5. Mann KG, Bovill EG, Krishnaswamy S. Surface-dependent reactions in the propagation phase of blood coagulation. *Ann N Y Acad Sci*. 1991;614: 63–75.
6. Krishnaswamy S, Field KA, Edgington TS, Morrissey JH, Mann KG. Role of the membrane surface in the activation of human coagulation factor X. *J Biol Chem*. 1992;267:26110–26120.
7. Smirnov MD, Ford DA, Esmon CT, Esmon NL. The effect of membrane composition on the hemostatic balance. *Biochemistry*. 1999;38: 3591–3598.
8. Jesty J, Lorenz A, Rodriquez J, Wun T. Initiation of the tissue factor pathway of coagulation in the presence of heparin: control by antithrombin III and tissue factor pathway inhibitor. *Blood*. 1996;87: 2301–2307.
9. van 't Veer C, Mann KG. Regulation of tissue factor initiated thrombin generation by the stoichiometric inhibitors tissue factor pathway inhibitor, antithrombin-III, and heparin cofactor-II. *J Biol Chem*. 1997;272: 4367–4377.
10. van 't Veer C, Golden NJ, Kalafatis M, Mann KG. Inhibitory mechanism of the protein C pathway on tissue factor-induced thrombin generation. *J Biol Chem*. 1997;272:7983–7994.
11. Toomey JR, Kratzer KE, Lasky NM, Stanton JJ, Broze GJ Jr. Targeted disruption of the murine tissue factor gene results in embryonic lethality. *Blood*. 1996;88:1583–1587.
12. Rosen ED, Chan JC, Idusogie E, Clotman F, Vlasuk G, Luther T, Jalbert LR, Albrecht S, Zhong L, Lissens A, Schoonjans L, Collen D, Castellino FJ, Carmeliet P. Mice lacking factor VII develop normally but suffer fatal perinatal bleeding. *Nature*. 1997;390:290–294.
13. Cui J, O'Shea KS, Purkayastha A, Saunders TL, Ginsburg D. Fatal haemorrhage and incomplete block to embryogenesis in mice lacking coagulation factor V. *Nature*. 1996;384:66–68.
14. Huang ZF, Higuchi D, Lasky N, Broze GJ Jr. Tissue factor pathway inhibitor gene disruption produces intrauterine lethality in mice. *Blood*. 1997;90:944–951.
15. Dewerchin M, Liang Z, Moons L, Carmeliet P, Castellino FJ, Collen D, Rosen ED. Blood coagulation factor X deficiency causes partial embryonic lethality and fatal neonatal bleeding in mice. *Thromb Haemost*. 2000;83:185–190.
16. Sun WY, Witte DP, Degen JL, Colbert MC, Burkart MC, Holmback K, Xiao Q, Bugge TH, Degen SJ. Prothrombin deficiency results in embryonic and neonatal lethality in mice. *Proc Natl Acad Sci U S A*. 1998;95:7597–7602.
17. Jalbert LR, Rosen ED, Moons L, Chan JC, Carmeliet P, Collen D, Castellino FJ. Inactivation of the gene for anticoagulant protein C causes lethal perinatal consumptive coagulopathy in mice. *J Clin Invest*. 1998; 102:1481–1488.

Downloaded from http://ahajournals.org by on March 25, 2026

18. Bi L, Lawler AM, Antonarakis SE, High KA, Gearhart JD, Kazazian HH Jr. Targeted disruption of the mouse factor VIII gene produces a model of hemophilia A. *Nat Genet*. 1995;10:119–121.
19. Kundu RK, Sangiorgi F, Wu LY, Kurachi K, Anderson WF, Maxson R, Gordon EM. Targeted inactivation of the coagulation factor IX gene causes hemophilia B in mice. *Blood*. 1998;92:168–174.
20. O'Marcaigh AS, Nicols WL, Hassinger NL, Mullins JD, Mallouh AA, Gilchrist GS, Owen WG. Genetic analysis and functional characterization of prothrombins Corpus Christi (Arg382-Cys) Dhahran (Arg271-His), and hypoprothrombinemia. *Blood*. 1996;88:2611–2618.
21. Henriksen RA, Mann KG. Substitution of valine for glycine-558 in the congenital dysthrombin Quick II alters primary substrate specificity. *Biochemistry*. 1989;28:2078–2082.
22. Bernardi F, Marchetti G, Patracchini P, Volinia S, Gemmati D, Simioni P, Girolami A. Partial gene deletion in a family with factor X deficiency. *Blood*. 1989;73:2123–2127.
23. Seeler RA. Parahemophilia. factor V deficiency. *Med Clin North Am*. 1972;56:119–125.
24. Arbini AA, Mannucci M, Bauer KA. A Thr^{359} Met mutation in factor VII of an individual with a hereditary deficiency causes defective secretion of the molecule. *Blood*. 1996;87:5085–5094.
25. Matsushita T, Kojima T, Emi N, Takahashi I, Saito H. Impaired human tissue factor-mediated in blood clotting factor VIINagoya (Arg304 224 Trp): evidence that a region in the catalytic domain of factor VII is important for the association with tissue factor. *J Biol Chem*. 1994;269: 7355–7363.
26. al-Monhiry H, Ehmann WC. Congenital afibrinogenemia. *Am J Hematol*. 1994;46:343–347.
27. Nemerson Y. Tissue factor and hemostasis. *Blood*. 1988;71:1–8.
28. Morrissey JH, Macik BG, Neuenschwander PF, Comp PC. Quantitation of activated factor VII levels in plasma using a tissue factor mutant selectively deficient in promoting factor VII activation. *Blood*. 1993;81: 734–744.
29. Giesen PL, Rauch U, Bohrmann B, Kling D, Roque M, Fallon JT, Badimon JJ, Himber J, Riederer MA, Nemerson Y. Blood-borne tissue factor: another view of thrombosis. *Proc Natl Acad Sci U S A*. 1999;96: 2311–2315.
30. Osterud B. The role of platelets in decrypting monocyte tissue factor. *Semin Hematol*. 2001;38(4 suppl 12):2–5.
31. Zillmann A, Luther T, Muller I, Kotzsch M, Spannagl M, Kauke T, Oelschlagel U, Zahler S, Engelmann B. Platelet-associated tissue factor contributes to the collagen-triggered activation of blood coagulation. *Biochem Biophys Res Commun*. 2001;281:603–609.
32. Santucci RA, Erlich J, Labriola J, Wilson M, Kao KJ, Kickler TS, Spillert C, Mackman N. Measurement of tissue factor activity in whole blood. *Thromb Haemost*. 2000;83:445–454.
33. Takahashi H, Satoh N, Wada K, Takakuwa E, Seki Y, Shibata A. Tissue factor in plasma of patients with disseminated intravascular coagulation. *Am J Hematol*. 1994;46:333–337.
34. Butenas S, Mann KG. Kinetics of human factor VII activation. *Biochemistry*. 1996;35:1904–1910.
35. Komiyama Y, Pedersen AH, Kisiel W. Proteolytic activation of human factors IX and X by recombinant human factor VIIa: effects of calcium, phospholipids, and tissue factor. *Biochemistry*. 1990;29:9418–9425.
36. Lawson JH, Butenas S, Ribarik N, Mann KG. Complex-dependent inhibition of factor VIIa by antithrombin III and heparin. *J Biol Chem*. 1993;268:767–770.
37. Lawson JH, Butenas S, Mann KG. The evaluation of complex-dependent alterations in human factor VIIa. *J Biol Chem*. 1992;267:4834–4843.
38. Butenas S, Ribarik N, Mann KG. Synthetic substrates for human factor VIIa and factor VIIa-tissue factor. *Biochemistry*. 1993;32:6531–6538.
39. Bom VJJ, Bertina RM. The contributions of Ca^{2+}, phospholipids and tissue factor apoprotein to the activation of human blood-coagulation factor X by activated factor VII. *Biochem J*. 1990;265:327–336.
40. Lawson JH, Mann KG. Cooperative activation of human factor IX by the human extrinsic pathway of blood coagulation. *J Biol Chem*. 1991;266: 11317–11327.
41. Butenas S, van 't Veer C, Mann KG. Evaluation of the initiation phase of blood coagulation using ultrasensitive assays for serine proteases. *J Biol Chem*. 1997;272:21527–21533.
42. Brummel KE, Paradis SG, Butenas S, Mann KG. Thrombin functions during tissue factor-induced blood coagulation. *Blood*. 2002;100: 148–152.
43. Pieters J, Lindhout T, Hemker HC. In situ-generated thrombin is the only enzyme that effectively activates factor VIII and factor V in thromboplastin-activated plasma. *Blood*. 1989;74:1021–1024.
44. Ahmad SS, Rawala-Sheikh R, Walsh PN. Components and assembly of the factor X activating complex. *Semin Thromb Hemost*. 1992;18: 311–323.
45. Olson ST, Bjork I, Shore JD. Kinetic characterization of heparin-catalyzed and uncatalyzed inhibition of blood coagulation proteinases by antithrombin. *Methods Enzymol*. 1993;222:525–559.
46. Girard TJ, Warren LA, Novotny WF, Likert KM, Brown SG, Miletich JP, Broze GJ Jr. Functional significance of the Kunitz-type inhibitory domains of lipoprotein-associated coagulation inhibitor. *Nature*. 1989; 338:518–520.
47. Rapaport SI. The extrinsic pathway inhibitor: a regulator of tissue factor-dependent blood coagulation. *Thromb Haemost*. 1991;66:6–15.
48. Esmon NL, Owen WG, Esmon CT. Isolation of a membrane-bound cofactor for thrombin-catalyzed activation of protein C. *J Biol Chem*. 1982;257:859–864.
49. Walker FJ, Sexton PW, Esmon CT. The inhibition of blood coagulation by activated protein C through the selective inactivation of activated factor V. *Biochim Biophys Acta*. 1979;571:333–342.
50. Suzuki K, Stenflo J, Dahlback B, Teodorsson B. Inactivation of human coagulation factor V by activated protein C. *J Biol Chem*. 1983;258: 1914–1920.
51. Eaton D, Rodriquez H, Vehar GA. Proteolytic processing of human factor VIII. Correlation of specific cleavages by thrombin, factor Xa, and activated protein C with activation and inactivation of factor VIII coagulation activity. *Biochemistry*. 1986;25:505–512.
52. Novotny WF, Brown SG, Miletich JP, Rader DJ, Broze GJ Jr. Plasma antigen levels of the lipoprotein-associated coagulation inhibitor in patient samples. *Blood*. 1991;78:387–393.
53. Kalafatis M, Rand MD, Mann KG. The mechanism of inactivation of human factor V and human factor Va by activated protein C. *J Biol Chem*. 1994;269:31869–31880.
54. Nesheim ME, Canfield WM, Kisiel W, Mann KG. Studies of the capacity of factor Xa to protect factor Va from inactivation by activated protein C. *J Biol Chem*. 1982;257:1443–1447.
55. Lawson JH, Kalafatis M, Stram S, Mann KG. A model for the tissue factor pathway to thrombin, I: an empirical study. *J Biol Chem*. 1994; 269:23357–23366.
56. van 't Veer C, Kalafatis M, Bertina RM, Simioni P, Mann KG. Increased tissue factor-initiated prothrombin activation as a result of the $Arg^{506} \rightarrow Gln$ mutation in factor V^{LEIDEN}. *J Biol Chem*. 1997;272: 20721–20729.
57. van 't Veer C, Golden NJ, Kalafatis M, Simioni P, Bertina RM, Mann KG. An in vitro analysis of the combination of hemophilia A and factor V^{LEIDEN}. *Blood*. 1997;90:3067–3072.
58. van 't Veer C, Butenas S, Golden NJ, Mann KG. Regulation of prothrombinase activity by protein S. *Thromb Haemost*. 1999;82:80–87.
59. Butenas S, van 't Veer C, Mann KG. "Normal" thrombin generation. *Blood*. 1999;94:2169–2178.
60. van 't Veer C, Golden NJ, Mann KG. Inhibition of thrombin generation by the zymogen factor VII: implications for the treatment of hemophilia A by factor VIIa. *Blood*. 2000;95:1330–1335.
61. Rand MD, Lock JB, van 't Veer C, Gaffney DP, Mann KG. Blood clotting in minimally altered whole blood. *Blood*. 1996;88:3432–3445.
62. Cawthern KM, van 't Veer C, Lock JB, DiLorenzo ME, Branda RF, Mann KG. Blood coagulation in hemophilia A and hemophilia C. *Blood*. 1998; 91:4581–4592.
63. Brummel KE, Butenas S, Mann KG. An integrated study of fibrinogen during blood coagulation. *J Biol Chem*. 1999;274:22862–22870.
64. Butenas S, Branda RF, van 't Veer C, Cawthern KM, Mann KG. Platelets and phospholipids in tissue factor-initiated thrombin generation. *Thromb Haemost*. 2001;86:660–667.
65. Holmes MB, Schneider DJ, Hayes MG, Sobel BE, Mann KG. Novel, bedside, tissue factor-dependent clotting assay permits improved assessment of combination antithrombotic and antiplatelet therapy. *Circulation*.;2000;102:2051–2057.
66. Butenas S, Cawthern KM, van 't Veer C, DiLorenzo ME, Lock JB, Mann KG. Antiplatelet agents in tissue factor-induced blood coagulation. *Blood*. 2001;97:2314–2322.
67. Butenas S, Brummel KE, Branda RF, Paradis SG, Mann KG. Mechanism of factor VIIa-dependent coagulation in hemophilia blood. *Blood*. 2002; 99:923–930.

Downloaded from http://ahajournals.org by on March 25, 2026

68. Brummel KE, Paradis SG, Branda RF, Mann KG. Oral anticoagulation thresholds. *Circulation*. 2001;104:2311–2317.
69. Jones KC, Mann KG. A model for the tissue factor pathway to thrombin. *J Biol Chem*. 1994;269:23367–23373.
70. Mann KG. How much factor V is enough? *Thromb Haemost*. 2000;83:3–4.
71. Hockin MF, Jones KC, Everse SJ, Mann KG. A model for the stoichiometric regulation of blood coagulation. *J Biol Chem*. 2002;277: 18322–18333.
72. Undas A, Brummel KE, Musial J, Mann KG, Szczeklik A. Simvastatin depresses blood clotting by inhibiting activation of prothrombin, factor V, and factor XIII and by enhancing factor Va inactivation. *Circulation*. 2001;103:2248–2253.
73. Undas A, Brummel KE, Musial J, Mann KG, Szczeklik A. Blood coagulation at the site of microvascular injury: effects of low-dose aspirin. *Blood*. 2001;98:2423–2431.
74. Undas A, Brummel KE, Musial J, Mann KG, Szczeklik A. Pl^{A2} polymorphism of β_3 integrins is associated with enhanced thrombin generation and impaired antithrombotic action of aspirin at the site of microvascular injury. *Circulation*. 2001;104:2666–2672.
75. Undas A, Wojciech JS, Brummel KE, Musial J, Mann KG, Szczeklik A. Aspirin and cardioprotective effects of the Val34Leu polymorphism of factor XIII. *Circulation*. In press.
76. Jenny RJ, Pittman DD, Toole JJ, Kriz RW, Aldape RA, Hewick RM, Kaufman RJ, Mann KG. Complete cDNA and derived amino acid sequence of human factor V. *Proc Natl Acad Sci U S A*. 1987;84: 4846–4850.
77. Suzuki K, Dahlback B, Stenflo J. Thrombin-catalyzed activation of human coagulation factor V. *J Biol Chem*. 1982;257:6556–6564.
78. Monkovic DD, Tracy PB. Activation of human factor V by factor Xa and thrombin. *Biochemistry*. 1990;29:1118–1128.
79. Mann KG, Kalafatis M. Factor V. A combination of Dr. Jeckyll and Mr. Hyde. *Blood*. In press.
80. Matthews JM, Rizza CR. Clinical features of clotting factor deficiencies. In: Biggs R, Rizza CR, eds. *Human Blood Coagulation, Haemostasis and Thrombosis*. Oxford: Blackwell Scientific Publications; 1984:119–169.
81. Bolton-Maggs PHB. Factor XI deficiency. *Baillieres Clin Hematol*. 1996;9: 355–368.
82. Ragni MV, Sinha D, Seaman F, Lewis JH, Spero JA, Walsh PN. Comparison of bleeding tendency, factor XI coagulant activity, and factor XI antigen in 25 factor XI-deficient kindreds. *Blood*. 1985;65:719–724.
83. Seligsohn U. Factor XI deficiency. *Thromb Haemost*. 1993;70:68–71.
84. Lipscomb MS, Walsh PN. Human platelets and factor XI. Localization in platelet membranes of factor XI-like activity and its functional distinction from plasma factor XI. *J Clin Invest*. 1979;63:1006–1014.
85. Gailani D, Broze GJ. Factor XI activation in a revised model of blood coagulation. *Science*. 1991;253:909–912.
86. Keularts IMLW, Zivelin A, Seligsohn U, Hemker HC, Beguin S. The role of factor XI in thrombin generation induced by low concentrations of tissue factor. *Thromb Haemost*. 2001;85:1060–1065.
87. Tracy PB, Eide LL, Bowie EJ, Mann KG. Radioimmunoassay of factor V in human plasma and platelets. *Blood*. 1982;60:59–63.
88. Kane WH, Lindhout MJ, Jackson CM, Majerus PW. Factor Va-dependent binding of factor Xa to human platelets. *J Biol Chem*. 1980;255:662–669.
89. Chesney CM, Pifer D, Colman RW. Subcellular localization and secretion of factor V from human platelets. *Proc Natl Acad Sci U S A*. 1981;78: 5180–5184.
90. Walsh PN. Factor XI. In: Colman RW, Hirsh J, Marder VJ, Clowes AW, George JN, eds. *Hemostasis and Thrombosis: Basic Principles & Clinical Practice*. Philadelphia, Pa: Lippincott Williams & Wilkins; 2001: 191–202.
91. Hu CJ, Baglia FA, Mills DC, Konkle BA, Walsh PN. Tissue-specific expression of functional platelet factor XI is independent of plasma factor XI expression. *Blood*. 1998;91:3800–3807.
92. Tracy PB, Giles AR, Mann KG, Eide LL, Hoogendorn H, Rivard GE. Factor V (Quebec): a bleeding diathesis associated with a qualitative platelet factor V deficiency. *J Clin Invest*. 1984;74:1221–1228.
93. Mosesson MW. The roles of fibrinogen and fibrin in hemostasis and thrombosis. *Semin Hematol*. 1992;29:177–188.
94. Blomback B, Hessel B, Hogg D, Therkildsen L. A two-step fibrinogen-fibrin transition in blood coagulation. *Nature*. 1978;275:501–505.
95. Lorand L, Chenoweth D, Domanik RA. Chain pairs in the crosslinking of fibrin. *Biochem Biophys Res Commun*. 1969;37:219–224.
96. Doolittle RF, Yang Z, Mochalkin I. Crystal structure studies on fibrinogen and fibrin. *Ann N Y Acad Sci*. 2001;936:31–43.
97. Bajzar L, Manuel R, Nesheim ME. Purification and characterization of TAFI, a thrombin-activable fibrinolysis inhibitor. *J Biol Chem*. 1995;270: 14477–14484.
98. Bajzar L, Morser J, Nesheim M. TAFI, or plasma procarboxypeptidase B, couples the coagulation and fibrinolytic cascades through the thrombin-thrombomodulin complex. *J Biol Chem*. 1996;271:16603–16608.
99. Wang W, Boffa MB, Bajzar L, Walker JB, Nesheim ME. A study of the mechanism of inhibition of fibrinolysis by activated thrombin-activable fibrinolysis inhibitor. *J Biol Chem*. 1998;273:27176–27181.

Downloaded from http://ahajournals.org by on March 25, 2026